

Assignment for Week 6

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Package Loading

```
library(ggplot2)
```

Assignment

- 1) Let E, F , and G be events. Find the expressions (using $\cap \cup$ and c) for the events that of E, F , and G
 - a) only F occurs,
 - b) both E and F but not G occur,
 - c) at least one event occurs,
 - d) at least two events occur,
 - e) all three events occur,
 - f) none occurs,
 - g) at most one occurs,
 - h) at most two occur.
- 2) 1.17 (p. 57)
- 3) 1.19 (p. 57)
- 4) 1.20 (a) (b) (p. 57)
- 5) 1.39 (p. 62)
- 6) 1.56 (a) (b) (p. 68)

7) Use R to simulate the following problems:

- (a) If three players are dealt five cards from a standard 52 card deck, how what is the probability that at least one person gets exactly two aces?
- (b) Suppose that n letters are placed randomly in n envelopes, what is the probability that $n - 1$ letters will be placed in the correct envelope?
- (c) Suppose that n letters are placed at random in n envelopes, if q_n is the probability that no letter is placed in the correct envelope, which of the following has the largest q_n ? $n = 10$, $n = 21$, $n = 53$, $n = 100$.
- (d) If the n letters are placed in n envelopes, what is the expected number of letters that go to the correct destination if $n = 10$, $n = 20$, $n = 50$?

8) Plot the pmf of each distribution in the following list, with at least one variation in the listed parameter(s), using either a histogram or with a custom `table` and a `persp` plot.

- bernoulli, p
- binomial, p
- Poisson, λ
- Geometric, p
- Hypergeometric, m, n
- negative binomial, p
- Multinomial, size, $p = (p_1, p_2)$

9) Using `str()` and the R help file and worksheet, describe in words each of the elements of the object `chisqt1`.

```
x <- c(89,37,30,28,2)
p <- c(.40,.20,.20,.15,.05)
chisqt1 <- chisq.test(x, p = p)
```